

Data Problem Statement - Healthcare Dataset (Group 3)

Dataset: Healthcare Dataset (prasad22, Kaggle) - 55,500 hospital admission records, 15 columns, one flat CSV.

Who needs this data

- Hospital administrators, for resource allocation and cost planning (staffing by admission urgency, room/bed utilization, cost trends by medical condition), and insurance analysts, for billing and claims accuracy across providers (Blue Cross, Medicare, Aetna, UnitedHealthcare, Cigna).

What decisions this data should support

- Predicting room and staffing demand by admission type (Urgent / Emergency / Elective) and medical condition
- Tracking billing trends and cost drivers by insurance provider and condition
- Identifying which conditions or admission types generate abnormal or inconclusive test results, to inform care quality review

What is broken now

- **534 duplicate admission records.** Full-row duplicates inflate patient volume and room/bed usage counts, which would directly corrupt any resource-planning metric an administrator pulls from this data.
- **Negative billing amounts** (minimum observed: -2,008.49, against a mean of ~25,539). A hospital bill cannot be negative; this is either a data entry error or an unlabeled refund/adjustment that needs a clear business rule before any financial reporting can trust this column.
- **Inconsistent name capitalization** in Name and Doctor (e.g. "Bobby JacksOn", "LesLie TErRy"). This breaks exact-match joins, groupings, and deduplication on these fields worsening problem #1, since case-inconsistent duplicates likely aren't even fully caught yet.